



United International University
School of Science and Engineering
Department of Computer Science and Engineering
 Mid Term Examination || Trimester - Fall-2025
Course Title: Calculus and Linear Algebra || Math -2183
 Total Marks: 30 || Time: 1 hour 30 minutes.

Answer all the questions in order

- Q1 (a)** The daily production cost (in taka) of two products A and B is given by $C(x, y) = 5x^2 + 4y^2 + 2xy - 34x - 22y + 100$, where x is the number of units of product A produced per day and y is the number of units of product B produced per day. Find the values of x and y that minimize the production cost. Verify your answer. [5]
- (b)** Find $\frac{dy}{dx}$ for the curve $x^3 + y^3 - 4xy = 0$. Hence find the gradient/slope of the curve at the point (2, 2). [3]
- Q2 (a)** Show that $y = 2 \sin 3x$ is the solution of the differential equation $\frac{d^2y}{dx^2} + 9y = 0$ [2]
- (b)** A curve y is such that $\frac{dy}{dx} = 3 - 2e^{-2x}$. Given that the curve is passing through the point (0, 2) i.e. $y(0) = 2$. Find the equation of the curve. [3]
- (c)** A rectangular field has a length (x) of 50 meters and a width (y) of 30 meters. During measurement, small errors occur such that the length is increased by 0.2 meters (dx) and the width is increased by 0.1 meters (dy). Using **the total differential**, estimate the approximate change in the area $A = f(x, y) = xy$ of the field, due to these small measurement errors. [2]
- Q3 (a)** A cup of hot tea is kept in a room where the temperature is 28°C . The rate at which the temperature of the tea decreases is given by the model $\frac{dT}{dt} = -k(T - 28), k > 0$. If the initial temperature of the tea is 90°C , Solve the differential equation to find the temperature $T(t)$ of the tea at any time t . [3]
- (b)** Given that $z = x^2 + y^2 + 1$ and $x = ue^v$ $y = u \ln v$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ [4]
- Q4 (a)** Solve the following differential equations [6]
- i. $\frac{dy}{dx} - 3y = 2x, y(0) = 1$
- ii. $(3x^2 + y)dx + (x + 2y)dy = 0$
- (b)** The following table shows the different values of $x, f(x)$ and its derivatives. Based on the information given in the table, determine the interval(s) over which the function is **increasing and decreasing**. [2]

x	-2	1
$f(x)$	7	-3
$f'(x)$	0	0
$f''(x)$	-ve	+ve